

Homework 1

Elena Amadei & Min Zhu

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Abstract

In this homework, we perform the 8 exercises provided in the pdf "hw1-2026.pdf".

Exercise 1

a

$$A = \begin{bmatrix} 1 & 2 & 2003 & 2005 \\ 2 & 2 & 2002 & 2004 \\ 3 & 2 & 2001 & 2003 \\ 4 & 7 & 7005 & 7012 \end{bmatrix}$$

Let the columns of A be denoted by $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4$. Observe that

$$\mathbf{c}_2 + \mathbf{c}_3 = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 7 \end{pmatrix} + \begin{pmatrix} 2003 \\ 2002 \\ 2001 \\ 7005 \end{pmatrix} = \begin{pmatrix} 2005 \\ 2004 \\ 2003 \\ 7012 \end{pmatrix} = \mathbf{c}_4.$$

Since the columns of A are linearly dependent, we conclude that

$$\text{rank}(A) \leq 3.$$

Next, we consider the first three columns $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$. Consider the submatrix formed by these columns and the first, second, and fourth rows:

$$A' = \begin{bmatrix} 1 & 2 & 2003 \\ 2 & 2 & 2002 \\ 4 & 7 & 7005 \end{bmatrix}.$$

A direct computation shows that

$$\det(A') = 10 \neq 0.$$

Thus, the columns $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ are linearly independent, which implies

$$\text{rank}(A) \geq 3.$$

With the two restrictions above, we conclude that

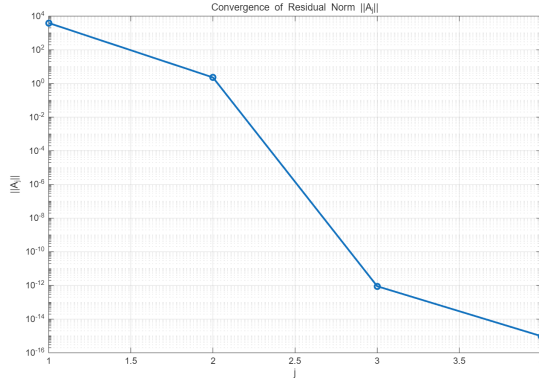
$$\text{rank}(A) = 3.$$

b

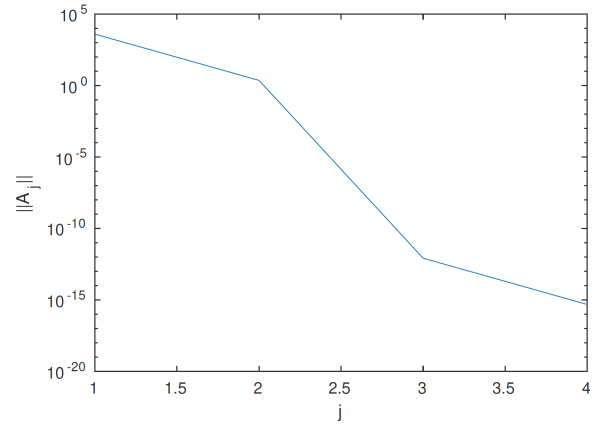
By applying Algorithm 1 (standard) to the matrix in (a), we obtain

$$\text{error after 3 interactions: } 8.9520 \times 10^{-13}.$$

It can be seen clearly in the margin plot [11c](#), when $j = 3$, $\|A_j\|$ reaches closely to 10^{-12} . The residual norm of A_j is shown as figure [1](#):



(a) error plot of q1.b



(b) image given in question

Figure 1: Image comparison

c

In this sub-question, we apply standard Algorithm 1 to the larger matrix and get the error plot.

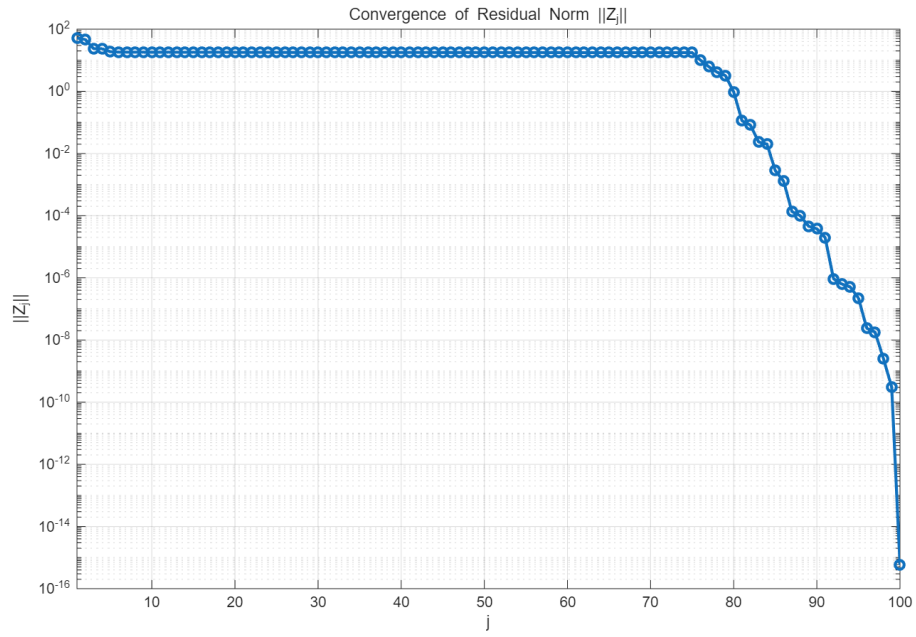


Figure 2: error plot of q1.c

d

In sub-question d, the greedy version of Algorithm 1 is used on the same larger matrix, corresponding error plot is shown in fig 3

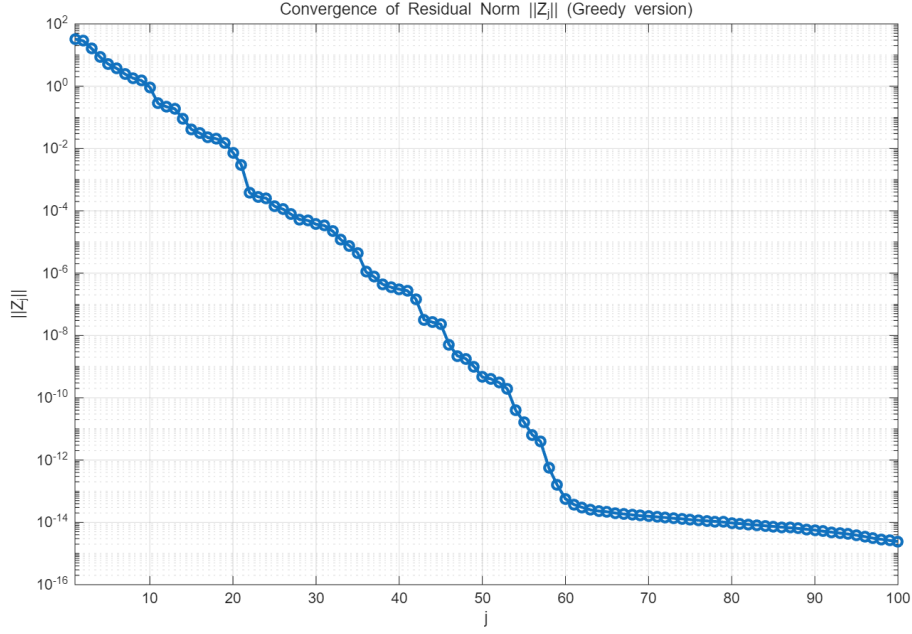


Figure 3: error plot of q1_d

Answer to the question: The greedy version of Algorithm 1 is better because, at each iteration, it selects the column with the largest norm from the remaining ones as the next basis vector. This strategy prioritizes the most informative direction at each step, resulting in a more efficient reduction of the approximation error. Therefore, we can obtain an approximation matrix that meets the prescribed accuracy with fewer iterations.

Exercise 2

Given a matrix A of dimension (2×2) , whose singular value decomposition exists as $A = USV^T$, we have

$$A = \begin{bmatrix} 5 & -1 \\ 5 & 7 \end{bmatrix}.$$

a

Given that S and V look as shown below, the scalar β and matrix U were asked to be found. In addition to that, a verification for the orthogonality of U was required.

$$S = \begin{bmatrix} \sqrt{80} & 0 \\ 0 & \sqrt{80} \end{bmatrix} \quad V = \beta \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

By substituting the S and V matrices into the formula $A = USV^T$, we had

$$\begin{bmatrix} 5 & -1 \\ 5 & 7 \end{bmatrix} = U \begin{bmatrix} \sqrt{80} & 0 \\ 0 & \sqrt{80} \end{bmatrix} \beta \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Since we know that U and V are both orthogonal and invertible, it followed that

$$I = V^T V = \beta^2 \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \beta^2 \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2\beta^2 I = 2\beta^2.$$

Therefore,

$$\beta = \pm \frac{1}{\sqrt{2}}.$$

Since, V is invertible and orthogonal either performing the computation with $-\beta$ or $+\beta$ will not affect the final conclusion. Hence, for the sake of simplicity, we performed the calculations with $+\beta = \frac{1}{\sqrt{2}}$. Consequently, one could observe that

$$A = USV^T \Rightarrow A = \sum_{i=1}^2 u_i \sigma_i v_i^T,$$

where $v_1^T = [\frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}}]^T$, $v_2^T = [-\frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}}]^T$, $\sigma_1 = \sqrt{80}$, and $\sigma_2 = \sqrt{20}$.

Hence, isolating the u vectors, we obtained

$$u_1 = \frac{Av_1^T}{\sigma_1} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}, u_2 = \frac{Av_2^T}{\sigma_2} = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \Rightarrow U = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}.$$

Finally, to check the orthogonality property, we checked that $U^T U = I$,

$$U = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ -\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

b

Here, the error $\|A - X\|$ had to be computed and compared with the theoretical definition. Given that $X = uv^T$, where u and v are vectors s.t. $X = \arg \min_{\text{rank}(X)=1} \|A - X\|$, it follows that the best rank-1 approximation is $X = \sigma_1 u_1 v_1^T$, where σ_1 , u_1 , v_1 are the first columns (or element) of the matrices S , U , and V .

Hence, let u and v^T from the definition of X be equal to $\sigma_1 u_1$ and v_1^T , respectively. Then,

$$X = uv^T = \sigma_1 u_1 v_1^T = 4\sqrt{5}\sqrt{10}^{-1}\sqrt{2}^{-1} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = 2 \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}.$$

The theoretical error of a rank- k approximation is $A_k = \|A - X\|_2 = \sigma_{k+1}$ by Eckhart-Young theorem,. Here, it would correspond to $\sigma_2 = \sqrt{20}$.

Computing the residual error explicitly, we had

$$X = \arg \min_{\text{rank}(X)=1} \left\| \begin{bmatrix} 5 & -1 \\ 5 & 7 \end{bmatrix} - \begin{bmatrix} 2 & 2 \\ 6 & 6 \end{bmatrix} \right\| = \arg \min_{\text{rank}(X)=1} \left\| \begin{bmatrix} 3 & -3 \\ -1 & 1 \end{bmatrix} \right\|.$$

It is to be noticed that the 2-norm is computed as the largest singular value, i.e. $\|M\|_2 = (M^T M)^{1/2} = \lambda$. Moreover, observing that $\sigma = \sqrt{\lambda}$, we performed the following computations to find the largest eigenvalue. Firstly,

$$(A - X)^T (A - X) = \begin{bmatrix} 3 & -1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & -10 \\ -10 & 10 \end{bmatrix}.$$

Consequently, we obtained $\lambda = 0$ and $\lambda = 20$ by the eigenvalues-eigenvectors relationship by computing

$$\begin{aligned} \det[(A - X) - \lambda I] &= 0 \\ \det \begin{bmatrix} 10 & -10 \\ -10 & 10 \end{bmatrix} &= 0 \\ \lambda(\lambda - 20) &= 0, \end{aligned}$$

Between the two values, $\lambda = 20$ was selected since it was the largest one. As a result, we ended up with

$$X = \arg \min_{\text{rank}(X)=1} \left\| \begin{bmatrix} 3 & -3 \\ -1 & 1 \end{bmatrix} \right\| = \sqrt{20} = \sigma_2,$$

which corresponds exactly to what the theory states.

Exercise 3

a

To obtain the lowest rank of the approximation matrix of A with an error less than 10^{-10} , we use the Eckart-Young theorem that,

$$X = \sum_{i=1}^k u_i v_i^T \sigma_i.$$

and the error is given by

$$\|V - X\| = \sigma_{k+1}.$$

So we plot the singular value decay curve and a horizon threshold of 10^{-10} to see the lowest rank k we can obtain.

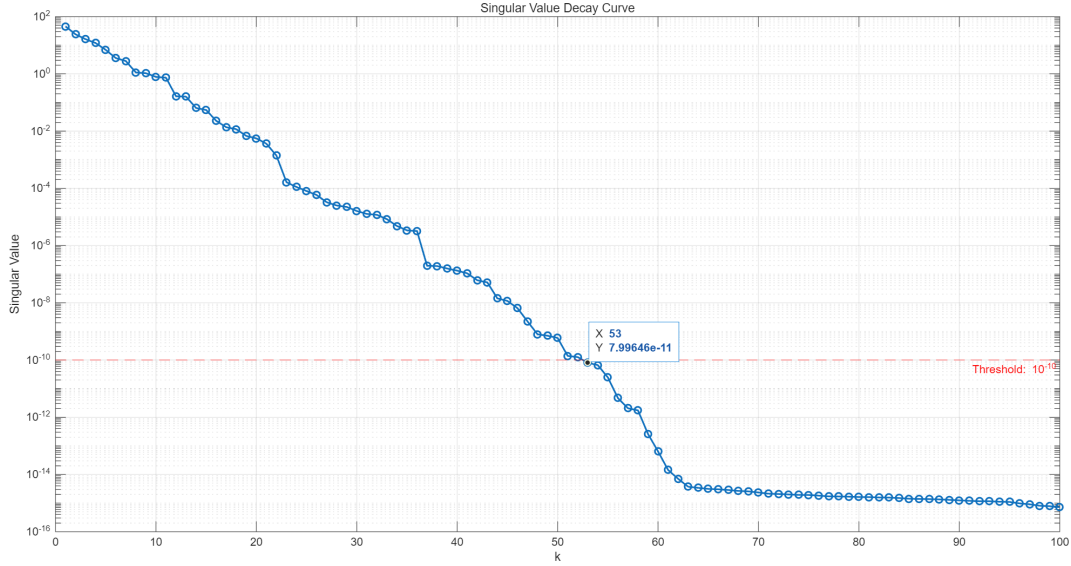


Figure 4: Singular Value Decay Curve

It shows that the minimum X under the threshold (a 2-norm error $\leq 10^{-10}$) is 53, thus $k + 1 = 53$, the lowest rank k is 52.

Answer to question sub a:

The lowest rank that approximates matrix A with an error 10^{-10} is 52.

b

The "SVD via Algorithm 1" first performs a QR factorization using the Gram-Schmidt method, and then applies the SVD to the smaller matrix R .

In our case, this process can be represented as:

$$A = Q_k R_k + A_k = Q_k \cdot (\hat{U} \Sigma V^T) + A_k = U \Sigma V^T + A_k$$

where $U = Q_k \hat{U}$. Therefore, $A_{approx} = U \Sigma V^T$ serves as our low rank approximation matrix.

Starting from a small rank, we sequentially increase k and compute the rank- k approximation as

$$A_{approx} = U(:, 1:k) \cdot S(1:k, 1:k) \cdot V(:, 1:k)',$$

until the corresponding approximation error falls below the threshold 10^{-10} .

Using this method, part of the result is shown in table 1.

Table 1: Results of SVD via Algorithm 1

Rank k	Approximation error (2-norm)
51	1.264×10^{-10}
52	7.996×10^{-11}
53	6.627×10^{-11}

Since the approximation error decreases monotonically as k increases, we conclude that the lowest rank we can obtain with "SVD via Algorithm 1" within error 10^{-10} is 52.

Exercise 4

- **Video quiz 1:** In this first video, the concept of low rank approximation was introduced. It was defined as the approximation given by Xy^T minimizing the error between the approximation and an original matrix A, i.e. $\min_{x,y} \|A - Xy^T\|$. An application of this method was also presented. For example, given the problem where we have a sequence of images, we may use low-rank approximation to find a (new) image that best match all original images ignoring scaling.
- **Video quiz 2a:** The video treated the QR-factorization from Gram-Schmidt standard procedure to obtain $V = QR$. Three main steps were described. They consist in the formula for obtaining the orthogonal basis w_k and q_k for the k-th column, the reformulation of the corresponding v_k , and finally the computation $V = QR$ with the elements just obtained, i.e.

$$V = [v_1 \quad \dots \quad v_p] = [q_1 \quad \dots \quad q_p] \begin{bmatrix} \|w_1\| & q_1^T v_2 & q_1^T v_3 & \dots & q_1^T v_p \\ 0 & \|w_2\| & q_2^T v_3 & \dots & q_2^T v_p \\ 0 & 0 & \|w_3\| & \dots & q_3^T v_p \\ 0 & 0 & 0 & \dots & \|w_p\| \end{bmatrix} = QR.$$

- **Video quiz 2b:** This video presented how to solve linear least squares problems via QR-factorization. After stating the definition of QR-factorization, it showed that the problem $Ax = b$ could be solved substituting $A = QR$ in the normal equation, i.e. $A^T Ax = A^T Ab = (QR)^T QRx = (QR)^T b$. This led to $Rx = Q^T b$, for invertible R , which corresponds to a triangular linear system that can be solved by backward substitution.
- **Video quiz 3a:** Singular value decomposition ($A = U\Sigma V^T$) is here introduced with relevant properties. Some examples can be the squared diagonal elements σ^2 of Σ are the eigenvalues of $A^T A$, and the norm of A and A^{-1} correspond to the largest eigenvalue $\|A\| = \sigma_1$ and the smallest in the inverse form $\|A^{-1}\| = \sigma_p^{-1}$, respectively.
- **Video quiz 3b:** Always about SVD, here the Eckart-Young theorem was explicitly defined. That is: the best rank-k approximation minimizing the residual error between a matrix A ($A = U\Sigma V^T$) and a matrix X of rank k is given by the first k singular vectors, i.e. $\operatorname{argmin}_{\operatorname{rank}(X)=k} \|A - X\| = \sum_{i=1}^k \sigma_i u_i v_i$. This theorem does not only give the minimizer, but provides also the error, i.e. $\|A - X\|_2 = \sigma_{k+1}$.
- **Video quiz 4a:** This video aimed defining concepts like index vector and permutation matrix with the corresponding properties. Examples of properties are (i) $A(:, [J, K]) = [A(:, J), A(:, K)]$, (ii) $(AB)(:, J) = A(B(:, J))$, and (iii) $(AB)(J, :) = A(J, :)B$ for J, K index vectors. Given these properties, it was possible derive that a permutation matrix can be obtain as $P = [I(:, J), I(:, \bar{J})]$.
- **Video quiz 4b:** The topic here was randomized SVD. Two main stages carry this procedure: while stage A consists in forming an approximate basis for the range of the matrix A, stage B restricts the matrix to the basis formed, performing accurate but expensive computations.
- **Video quiz 5:** Column ID factorization was here introduced. It allows for $A = CZ$, where C is a selection of columns of the original matrix A, and Z is a matrix containing expansion coefficients. This factorization can be used for rank approximation, but provides results at most as good as the approximation performed with SVD.

Exercise 5

c

We name the result of $A - uv^T$ as **E**. As we load the pictures and reshape each of them into a column, these columns j in **E** represents the difference between the j -th frame and the first frame.

$\|A - uv^T\|$ is the norm of matrix **E**, and its 2-norm value is 1.769×10^4 , its frobenius norm value is 2.453×10^4 . When 2-norm is used, it reflects the maximum deviation of all frames; When the Frobenius norm is applied, it represents the total deviation of the entire video from the first frame.

d

Here, the 2-norm value (1.769×10^4) is adopted as the tolerance for the approximation error. The same procedure in Task 3(b) is then applied, which consists of performing a QR factorization followed by an SVD approximation of the matrix R .

The experimental results indicate that, under this tolerance, the numerical rank of **A** is 1. To confirm this, we compute the rank-1 approximation

$$A_{\text{approx}} = U_{(:,1:n_{\text{rank}})} S_{(1:n_{\text{rank}}, 1:n_{\text{rank}})} V_{(:,1:n_{\text{rank}})}^T,$$

and evaluate the corresponding approximation error

$$Err = \|A - A_{\text{approx}}\|.$$

The computed error is $Err = 3.566 \times 10^3$, which is below the prescribed tolerance (1.769×10^4), thereby confirming that the numerical rank is indeed 1.

e

First, we compare the speed. The computational times required to decompose **A** using MATLAB's built-in svd function and the partial SVD method proposed in Task (d) are summarized in the table 2 below.

Table 2: Comparison of time on different SVD

Time of Matlab's SVD	Time of our partial SVD
1.0545 sec	5.0467 sec

It shows that Matlab's svd is much faster than our method. The slow performance of our method may be because our algorithm uses the Gram-Schmidt process for QR factorization, which is generally less efficient than the Householder-based QR employed in Matlab.

Next, we compare the rank-1 approximation error.

Table 3: Comparison of error on different approximation method

Method	Brief description	Error
Method in (c)	simple approximation by copy the first frame	2.45×10^4
Method in (d)	partial svd (first qr, then do svd on R)	1.73×10^4
Matlab's svd	Matlab's economy version of svd	1.73×10^4

The result shows that, Method in (d) and Matlab's svd have the same error, both lower than the simple approximation method in (c). This outcome is reasonable in theory because, method in (c) simply replicates the first frame and therefore fail to capture other structural information of the matrix. In contrast, both method in (d) and Matlab's svd are derived from performing SVD directly of the original matrix and are thus equivalent in terms of approximation quality. According to the Eckart-Young Theorem, these two methods yield the optimal rank-1 approximation, both achieving minimized error.

Exercise 6

In this exercise, we performed the analysis of the video "roundabout.mkv" and the corresponding frames. We stored the snapshots of the video in a matrix V of dimensions 1080264×56 .

a

In this first task, we performed the greedy version of Algorithm 1 up to 20 iterations and we computed the residual error per-iteration, i.e. $\|A_j\| = \|V - Q_k R_k\|$ for $j = 1, \dots, 20$. This estimate measures the decay of the residual error of the rank- k approximation at each iteration. It has been computed using the Frobenius norm, which reduces to the 2-norm since each column corresponds to a vector. One can notice that with only 20 iterations the error dropped pretty fast reaching almost zero.

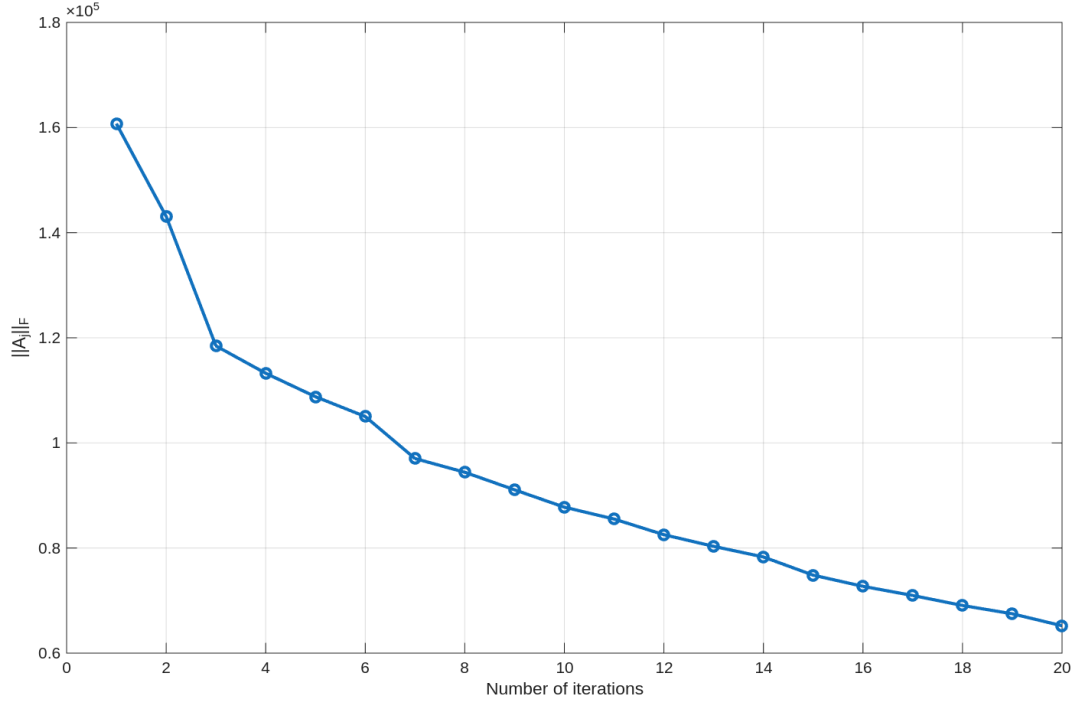


Figure 5: Residual error decay at each distribution $j = 1, \dots, 20$.

b

The following figure presents the errors at individual snapshots for both a rank-10 and rank-20 approximation matrix. The same measure as in **task a** has been computed. In the plot it can be seen that exactly 10 and 20 points lay on the x-axis at zero. This shows how the algorithm selected 10 and 20 columns respectively as on-basis for the range of A . Further observing the plot, whenever a spike is showed, it means that a higher error is registered for that frame. This may mean that the frame contains distinct information that are not captured by the current frame approximation. On the contrary, low points correspond to low error areas, which means that the frames look like the selected basis". In addition to that, it is worth noticing that the first 10 points are the same for both of the 2 low rank approximations. Comparing the two curves, a reduction of the residual error can be observed for the frames that were not selected as basis for the approximation, with the orange line (rank-20 approximation) being consistently lower than the blue line (rank-10 approximation). Therefore, the rank-20 approximation captures more variability and more details.

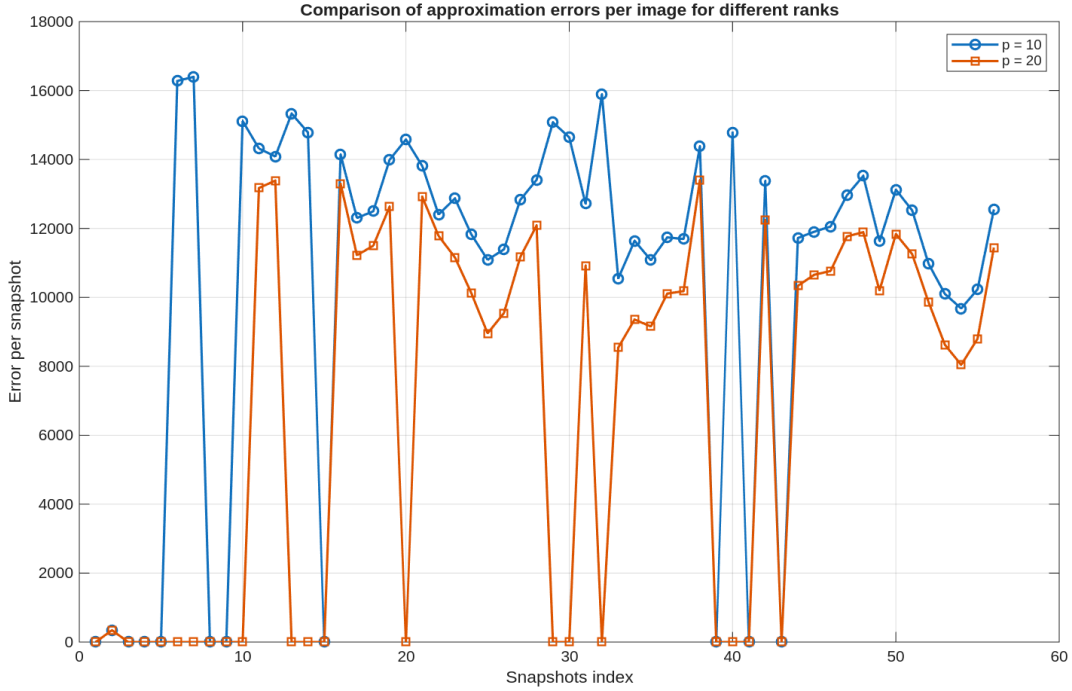


Figure 6: Comparison of residual error per-snapshot in a rank-10 and rank-20 approximation.

c

An SVD approximation from the QR rank-20 approximation has been performed to compute additional rank-1, rank-2 and rank-5 approximations. The resulting images produced by each of them can be observed below.



(a) Image from a SVD rank-1 approximation.



(b) Image from a SVD rank-2 approximation.



(c) Image from a SVD rank-5 approximation.

Figure 7: Image comparison

d

Here, we recreated two videos using the SVD low-rank approximation with rank = 1 and rank = 5 using the "makevideo" program provided.

Exercise 7

a

In this question, we apply randomized SVD on matrix in task 6. With target rank $k=5$ fixed, a curve of approximation error along oversampling parameter number s from 1 to 25 is shown in Fig 8.

In this question, we apply the randomized SVD algorithm to the matrix considered in Task 6. With the target rank fixed at 5, we investigate the effect of the oversampling parameter s on the

approximation error. Figure 8 illustrates the variation of the approximation error as s increases from 0 to 25.

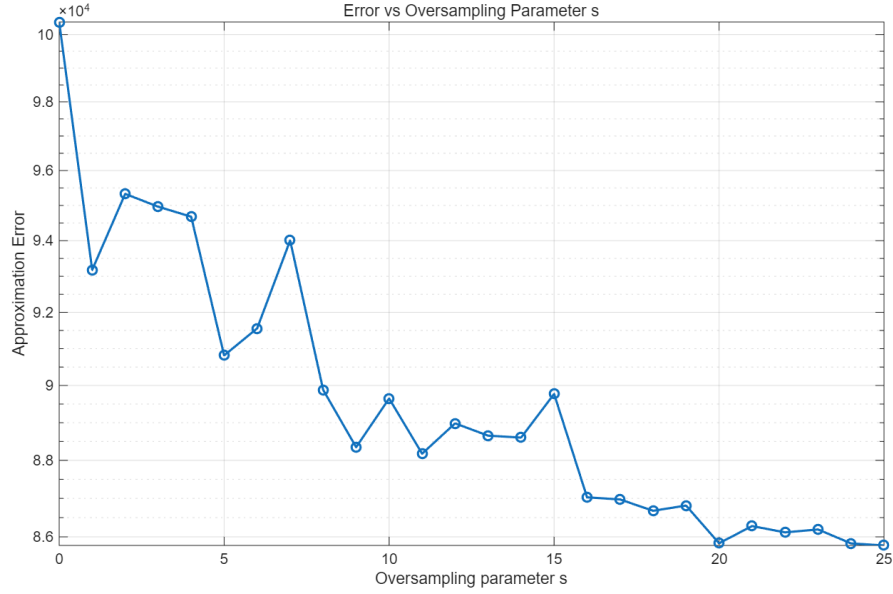


Figure 8: Approximation error to S

b

To estimate the average CPU time of the three methods, we repeatedly apply each method to compute a low-rank approximation of A . Specifically, we conduct 30 independent trials for each method, recording both the computational time required to obtain the approximate matrix and the corresponding approximation error in each run. The reported results are obtained by averaging over these repeated experiments as in 4.

Table 4: Comparison of Execution Time and Error Across Different SVD Methods

Method	Avg Time (s)	Avg Error
Random_SVD	0.446614	8.7944×10^4
Greedy QR	2.450405	1.0878×10^5
Matlab SVD	1.295089	8.3397×10^4

Exercise 8

As preliminary step in this exercises, it was asked to complete the ID skeleton code provided. Here we provide the completed version of the ID_col.m function:

```
function [C,Z]=ID_col(A, kk)

[Q,R,P]=qr(A);
Qs=Q(:, 1:kk);
Rs=R(1:kk,:);

R11=Rs(:, 1:kk);
R12=Rs(:, (kk+1):length(A));

C=A*P(:, 1:kk);
I=eye(kk, kk);
```

```
Z= [I, inv(R11)*R12]*P';
```

a

A comparison of the relative errors produced by the best rank-k approximation and the ID-approximation is provided in the following plot. Since the best rank-k approximation was asked for the first case, we computed an SVD-approximation, which indeed provides the best results in terms of quality although being very computationally expensive. Furthermore, we concluded that the 25% of accuracy, computed as $\frac{\|A-\hat{A}\|}{\|A\|} = 0.25$, can be obtained at rank 126 for the SVD-approximation and 768 for the ID-approximation.

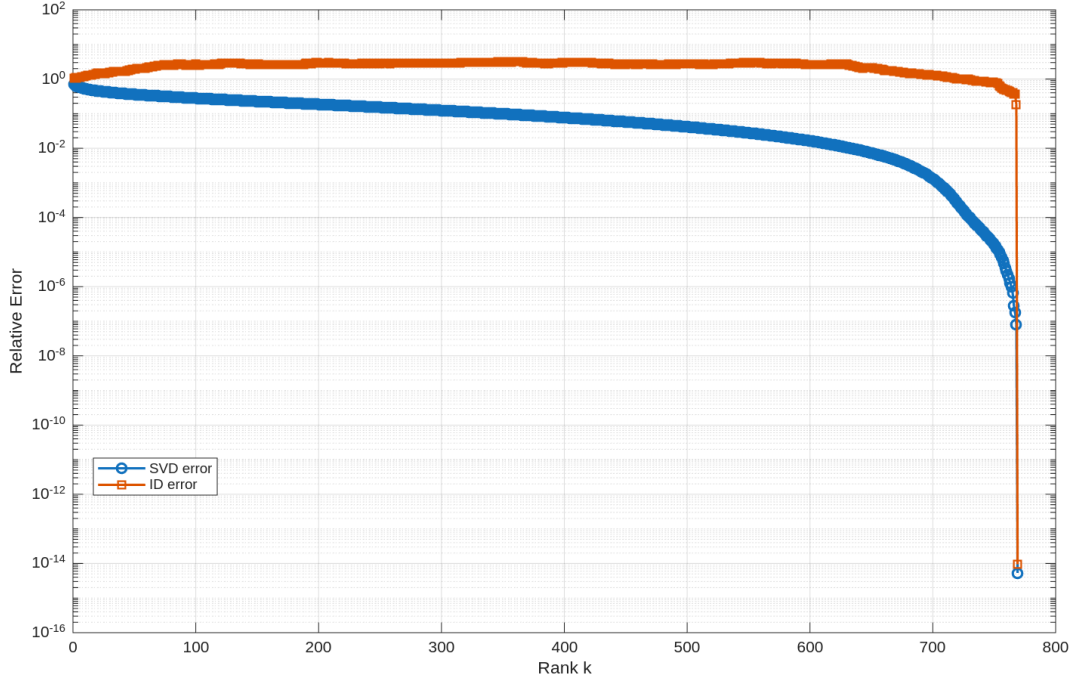
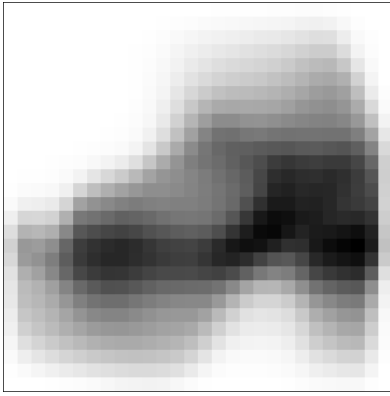


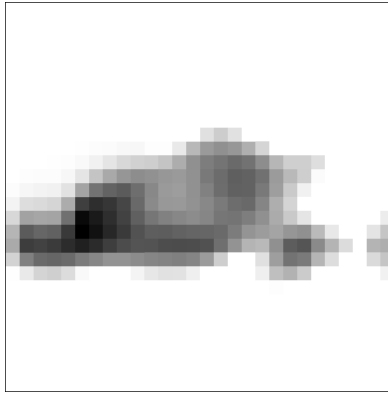
Figure 9: Semilog plot for the residual error from a SVD-approximation and an ID-approximation.

b

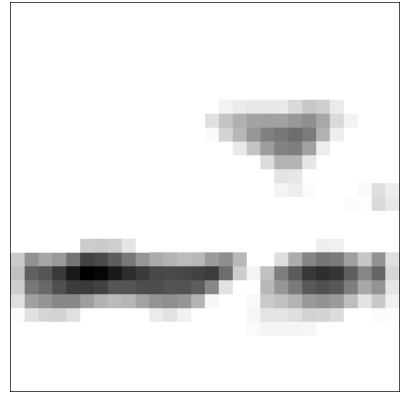
Here, we present the images corresponding to the first three singular components, and first three interpolator decomposition vectors. Comparing the two approximation methods, we know that SVD always provides the solution with the lowest possible error, but produces values that could not be useful and applicable to real-life contexts. As in this case, it can be seen that the objects in the plots from the singular vectors are barely identifiable. On the other hand, ID-approximation does not always return the best solution, nevertheless it conserves the structure's patterns of the original matrix allowing for a good reproducibility.



(a) u_1 singular vector.



(b) u_2 singular vector.

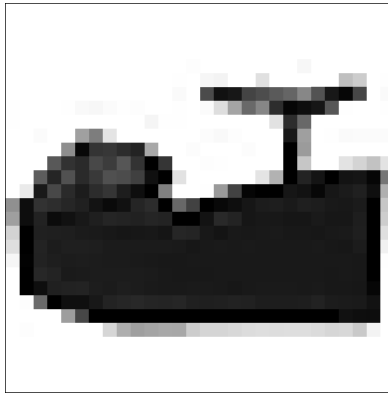


(c) u_3 singular vector.

Figure 10: First three singular vectors corresponding to u_1, u_2, u_3 .



(a) c_1 ID vector.



(b) c_2 ID vector.



(c) c_3 ID vector.

Figure 11: First three interpolatory decomposition vectors corresponding to u_1, u_2, u_3 .